available at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/agwat

Assessment of winter wheat production under early sowing with supplemental irrigation in a cold highland environment using CropSyst simulation model

B. Benli [a,*], M. Pala [a], C. Stockle [b], T. Oweis [a]

[a] International Center for Agricultural Research in the Dry Areas (ICARDA), P.O Box 5466, Aleppo, Syria
[b] Washington State University, Pullman, WA 99164-6120, USA

ARTICLE INFO

Article history:
Received 16 October 2006
Accepted 13 June 2007
Published on line 5 September 2007

Keywords:
CropSyst model
Supplemental irrigation
Bread wheat
Early sowing
Highland
Turkey

ABSTRACT

The most important question in supplemental irrigation management is when and how much water to apply. It is a laborious and expensive task to develop supplemental irrigation schedules solely by conventional field experimentation. A cropping system simulation model (CropSyst) was evaluated for its ability to simulate growth, biomass, grain yield and evapotranspiration (ET) of wheat sown early with supplemental irrigation (SI). This was computed under rainfed conditions in a semiarid climate with cold winters in the highlands of Turkey. Experimental data from three growing seasons (1998–2001) were used. The experimental design incorporates Bezostia bread wheat cultivar tested under two main treatments: no irrigation at sowing (rainfed-A_1) and irrigation at sowing with 50 mm of water (A_2); and two sub treatments: rainfed (B_1) and spring supplemental irrigation to replenish the total water requirement at 0–90 cm soil profile (B_2) at the Ankara Research Institute of Rural Services. Crop input parameters were selected from the model documentation and experimental data. A few cultivar-specific parameters were adjusted within a narrow range of typical fluctuations by model calibration. Results showed that CropSyst was able to simulate yield, biomass and evapotranspiration as observed in the field experiments. Overall, the Willmott Index of agreement between simulated and observed values of grain yield, biomass and ET were 0.98, 0.76 and 0.91, respectively. CropSyst model predicted better the seasonal evapotranspiration under full supplemental irrigated conditions (A_2B_2) than under rainfed conditions (A_1B_1), with values of the Willmott index of agreement being 0.97 and 0.89, respectively. The model was run for 20 years (1982–2001) including the 4-year experimental period. Data showed that wheat grain yield could be improved by 15, 19 and 25% with applying only 50 mm of water at the sowing time of 15 October, 1 October and 15 September, respectively. In 80% of the cases, the respective SI applications would give 2.75, 2.7 and 2.95 t ha^{-1}, of the long-term average rainfed yield of 2.1 t ha^{-1}, respectively.

© 2007 Elsevier B.V. All rights reserved.

1. Introduction

Central Anatolia Plateau (CAP) is a typical rainfed area. It covers around 10 million hectares of cultivated land and wheat is the major crop. Annual rainfall varies from 250 mm to 500 mm and is below wheat water requirement. Wheat yield, as a result of inadequate rainfall and its suboptimal distribution is low and ranges from 0.9 to 2.5 t ha^{-1} with an average of

* *Corresponding author*. Tel.: +90 532 483 7340; fax: +90 312 496 1465.
E-mail address: bogachanbenli@gmail.com (B. Benli).
0378-3774/$ – see front matter © 2007 Elsevier B.V. All rights reserved.
doi:10.1016/j.agwat.2007.06.014

less than 2.0 t ha^{-1}. This is way below the potential yield of over 6 t ha^{-1}.

Supplemental irrigation, i.e., applying limited amounts of water to alleviate drought spells in the raining season, is one way to increase productivity. To improve yield, it is necessary to integrate supplemental irrigation strategy into a comprehensive cropping system approach. However, in CAP region, both surface and groundwater resources are limited. Groundwater is used when the surface water is not sufficient, and mostly in remote areas. Depth to ground water ranges from 80 to 150 m for deep aquifers. Available information shows majority of the farmers (65%) in CAP consider their irrigation applications insufficient for maximizing wheat yield (Oweis et al., 2001). Perrier and Salkini (1991) indicated, however, that farmers tend to over irrigate and that the return for the water in terms of grain is low due to lack of proper scheduling and optimal management of irrigation water. Therefore, the most important question in SI management in CAP region is determining when and how much water to apply (Oweis, 1997). For example, Ilbeyi et al. (2006), indicated that, when early rain was inadequate for crop germination, SI, given at sowing substantially increased wheat yield by more than 65%, adding about 2.0 t ha^{-1} to the average rainfed yield of 3.2 t ha^{-1} at the highlands of Central Anatolia, Turkey. Plants emerging early in autumn, grow more vigorously and develop faster in the following spring than plants which emerge late as subject to frost, leading to higher yields with higher water productivity. Similarly, research results from highlands of Iran has shown that, 1 t ha^{-1} of rainfed yield would be increased to 2.5, 3.0 and 3.5 t ha^{-1} by 1/3, 2/3 and full supplemental irrigation, respectively (Tavakkoli and Oweis, 2004). However, these studies are laborious and expensive tasks to be undertaken by conventional field experimentations. Therefore, crop simulation models are effectively used for filling the gap and outscaling the results.

Cropping system simulation models can be used to predict the effect of weather, soil properties, plant characteristics and management practices on the soil water balance, nutrient dynamics and growth of crops. Therefore, they can enhance our understanding of cropping systems performance under different water and nitrogen regimes. Models may also be used to assess the effects of management practices and plant characteristics on crop performance over a period that is long enough to characterize the climatic variability of a site (van Keulen and Seligman, 1987), leading to improvements in the efficacy of decision-making for fertilizer and water management. However, suitable field experiments are required for model validation, a necessary step before model applications can be developed for a given region (Cabelguenne et al., 1990; Kropff et al., 1994; Lengnick and Fox, 1994).

Cropping Systems Simulation Model (CropSyst) represents an effort to simulate the growth of single crops or crop rotations in response to weather/soil/management scenarios and provide an estimate of environmental impact (Stockle and Nelson, 1994; Stockle et al., 1994). CropSyst is a multi-year and multi-crop daily time step simulation model. It has been developed to serve as an analytical tool to study the effect of cropping systems management on productivity and the environment. The model simulates the soil water budget, soil–plant nitrogen budget, crop canopy and root growth, dry matter production, grain yield, residue accumulation and decomposition and erosion. Management options include cultivar selection, crop rotation, irrigation, nitrogen fertilization, tillage operations and residue management.

The water budget in the model includes rainfall, irrigation, runoff, interception, water infiltration and redistribution in the soil profile, crop transpiration, and evaporation. The nitrogen budget in CropSyst includes nitrogen application, nitrogen transport, nitrogen transformations, ammonium sorption and crop nitrogen uptake. The calculation of daily crop growth, expressed as biomass increase per unit area, is based on a minimum of four limiting factors, namely light, temperature, water, and nitrogen. Pala et al. (1996) suggested that minor adjustments of some of these parameters, accounting for cultivar-specific differences, are desirable whenever suitable experimental information is available. Details on the technical aspects and use of the CropSyst model have been reported elsewhere (Stockle et al., 1994; Stockle and Nelson, 1994).

The objective of this study was to evaluate options for improved winter wheat production with early sowing by introducing SI in a semi-arid climate with cold winters and cool springs in the highlands using CropSyst simulation model.

2. Material and methods

2.1. *The experiment*

The data used for model validation was obtained from a field experiment conducted at the Ankara Research Institute of Rural Services (ARIRS) (30°53′N; 32°45′E; elevation: 924 m) during the 1998/1999, 1999/2000 and 2000/2001 seasons (Ilbeyi et al., 2006). The experiment was not specifically designed to

Table 1 – Summary of pre-plant soil test results for ARIRS experimental terrain

Year	Depth (cm)	Total salinity (%)	pH	Lime (%)	Organic matter (%)	Available (kg/ha)	
						P_2O_5	K_2O
1998	0–20	0.07	7.9	28.20	1.66	35.6	744.9
	20–40	0.06	8.0	27.98	1.64	24.4	588.9
1999	0–20	0.08	7.9	24.97	1.64	42.7	807.3
	20–40	0.08	8.0	26.91	1.56	46.1	791.8
2000	0–20	0.09	7.5	28.91	1.45	56.2	772.2
	20–40	0.07	7.7	28.54	1.37	46.8	620.1

provide data for validation of CropSyst. Therefore, some data, such as the seasonal evolution of leaf area index and biomass, are missing. The experiment was designed as a split-block with three replications. Plot dimensions were 27.12 m^2 (4.52 m × 6.0 m) at sowing and 11.70 m^2 at harvest in order to avoid side effects. The experimental design incorporates Bezostia bread wheat cultivar tested under two main treatments: no irrigation at sowing (A_1) and irrigation at sowing with 50 mm of water (A_2); and two sub-treatments in spring as rainfed (B_1) and irrigation to replenish the total water requirement at 0–90 cm of maximum effective root zone (B_2). Neutron probe access tubes were installed to a depth of 120 cm in one replication plot of each treatment/sub-treatment combination, and soil water content was monitored weekly. Evapotranspiration was calculated using the mass balance equation given by Raats and Warrick (1983)

$$ET = P + NI \pm \Delta S \quad (1)$$

where P is precipitation, NI the depth of water needed to refill the root zone for the full SI treatment using the current root zone depth with 50% allowable depletion and ΔS is the change in soil water storage during the measurement period.

Winter wheat was sown in October of each year by drill, with 13.3 cm row spacing and harvested in mid-July. During sowing, 75 kg N ha^{-1} and 75 kg P_2O_5 ha^{-1} of fertilizers were applied and an additional 75 kg N ha^{-1} was applied in March each year. Soil samples at 0–20 and 20–40 cm depth were taken at planting from one replication plot of each treatment/sub-treatment combination in each year (Table 1).

The soil of the experimental site is brown clay-loam with pH ranging from 7.5 to 8.0 and a composition of 36% sand, 26% silt and 38% clay. Soil moisture at field capacity and at wilting point was 35% and 20% by volume, respectively. Mean soil specific gravity was 1.24, with infiltration rate of 20 mm h^{-1}. Electrical conductivity of the saturated soil paste (EC_e) ranged from 0.75 to 2.50 dS m^{-1}, lime from 24.97 to 29.70% and organic matter from 1.37 to 1.66%. Available potassium and phosphorus values ranged from 744.9 kg ha^{-1} to 807.3 kg ha^{-1} and 35.6 kg ha^{-1} to 56.2 kg ha^{-1} at 0–20 cm depth and 588.9 kg ha^{-1} to 791.8 kg ha^{-1} and 24.4 kg ha^{-1} to 46.8 kg ha^{-1} at 20–40 cm depth, respectively.

The weather data from a meteorological station located near the experimental site was used. Precipitation, max and min temperatures, air humidity, wind speed, and sunshine hours were measured daily. Monthly rainfall and max and min temperature data for each growing season are summarized in Table 2.

2.2. Calibration and validation

Total biomass, grain yield, total and seasonal evapotranspiration (ET), computed from the soil-moisture measurements from the treatments A_1B_1 and A_2B_2 of the 1998/1999 season, which had rainfall above the long-term average and well distributed over the season, were used for model calibration. The values of the crop input parameters (Table 3) were either taken from the CropSyst manual (Stockle and Nelson, 1994) or set to the values observed in the experiments. The calibration consisted of slight adjustments

Table 2 – Summary of climate data for each season of the experimental period and the long-term averages (1963–2001)

Month	1998–1999			1999–2000			2000–2001			Long-term average		
	Precipitation (mm)	Mean daily maximum temperature (°C)	Mean daily minimum temperature (°C)	Precipitation (mm)	Mean daily maximum temperature (°C)	Mean daily minimum temperature (°C)	Precipitation (mm)	Mean daily maximum temperature (°C)	Mean daily minimum temperature (°C)	Precipitation (mm)	Mean daily maximum temperature (°C)	Mean daily minimum temperature (°C)
October	24.8	21.6	8.5	36.3	19.6	8.5	5.0	25.6	11.8	27.2	18.8	7.4
November	30.5	13.1	4.8	27.5	12.2	2.0	23.4	18.1	6.6	35.1	11.5	2.4
December	40.8	7.1	1.8	38.8	10.0	0.7	17.1	15.3	3.2	47.2	5.4	0.9
January	33.1	6.3	−0.2	46.5	−0.4	−6.6	35.5	5.0	−1.3	39.3	3.0	3.3
February	74.1	6.9	−0.6	53.6	2.0	−4.9	2.2	6.6	−0.6	31.4	5.4	2.4
March	41.4	11.5	1.2	36.0	9.0	1.4	37.1	8.4	−0.6	36.5	10.8	1.0
April	16.5	17.0	5.8	78.8	17.8	7.4	26.9	16.5	5.6	48.3	16.1	5.5
May	16.3	22.8	9.2	51.0	20.7	9.0	28.8	17.8	6.1	50.5	20.9	9.5
June	39.6	25.9	13.5	33.2	25.5	12.2	91.3	20.0	8.5	31.5	25.6	12.7
July	60.9	30.6	16.5	–	33.3	17.4	0.1	28.2	13.2	15.6	29.2	15.5
August	37.0	29.6	16.8	9.4	28.8	15.2	8.2	32.8	18.1	11.3	28.8	15.4
September	20.0	25.3	11.6	5.0	25.6	11.8	6.6	30.7	17.3	13.0	24.9	11.8
Total/Mean	435.0	18.1	7.4	416.1	17.0	6.2	282.2	18.8	7.3	386.8	16.7	6.2

Table 3 – Crop parameters of Bezostia wheat cultivar

Parameters	Activity	Bezostia
Growth		
Above-ground biomass-transpiration coefficient (kg/m^3)	Calibration	4.10
Light to above-ground biomass conversion (g/MJ)	Default	2.80
Actual to potential transpiration ratio that limits leaf area grow	Default	0.80
Actual to potential transpiration ratio that limits root growth	Default	0.20
Optimum mean daily temperature for growth (°C)	Default	15.00
Maximum water uptake (mm/day)	Default	14.00
Leaf water potential at the onset of stomatal closure (−J/kg)	Calibration	−1300.00
Wilting leaf water potential (−J/kg)	Default	−2000.00
Morphology		
Maximum rooting depth (m)	Observed	1.80
Initial green leaf area index (m^2/m^2)	Default	0.05
Maximum expected leaf area index (LAI) (m^2/m^2)	Default	5.00
Fraction of maximum LAI at physiological maturity	Default	0.80
Specific leaf area (m^2/kg)	Calibration	22.00
Stem/leaf partition coefficient	Calibration	4.00
Leaf duration (degree-days)	Default	950.00
Extinction coefficient for solar radiation	Default	0.48
Leaf duration sensitivity to water stress	Default	0.00
ET crop coefficient at full canopy	Default	1.20
Phenology		
Degree days emergence (°C-days)	Observed	210.00
Degree days flowering (°C-days)	Observed	755.00
Degree days grain filling (°C-days)	Observed	900.00
Degree days physiological maturity (°C-days)	Observed	1270.00
Base temperature (°C)	Default	0
Cut-off temperature (°C)	Default	25.00
Phenologic sensitivity to water stress	Default	0.50
Harvest		
HI sensitivity to water stress during flowering	Calibration	0.15
HI sensitivity to water stress during grain filling	Calibration	0.15
Translocation factor	Default	0.30

of selected crop input parameters to reflect reasonable simulations. These adjustments were around values that were either typical for the crop species or known from previous experiences with the model.

The model was subsequently validated for the site conditions using two treatments (A_1B_2 and A_2B_1) for the 1998/1999 season and four (A_1B_1, A_1B_2, A_2B_1 and A_2B_2) for the 1999/2000 and 2000/2001 growing seasons and using the crop model parameter values calibrated as mentioned above with associated water management. Soil characteristics, initial conditions of available soil moisture, nitrogen and organic matter and daily weather data were input into CropSyst as observed in the experiments.

Model evaluation was conventionally made by comparing simulation outputs with observed and collected data. In these evaluations, two types of different measures were used. Average difference is alternatively described by the root mean square errors (RMSE) and expressed as:

$$\mathrm{RMSE} = \left[N^{-1}\sum_{i=1}^{N}(P_i - O_i)^2\right]^{0.5} \quad (2)$$

where N is the number of cases, O the observed and P is the predicted variate.

Other indicators of performance was Willmott index of agreement (d) that takes on values from 0 to 1.0, with an index of 1.0 indicating perfect agreement (Willmott, 1982) and expressed as:

$$d = 1 - \left[\frac{\sum_{i=1}^{N}(P_i - O_i)^2}{\sum_{i=1}^{N}(|P_i| + |O_i|)^2}\right] \quad 0 \le d \le 1 \quad (3)$$

2.3. *Model application scenarios*

As we have obtained promising results from the model calibration and validation on the 4-year experimental period, we have applied different scenarios using 20 years (1982–2001) climate data for the same site. Scenarios included various sowing dates such as 15 September (ESI_1), 1 October (ESI_2) and 15 October (ESI_3) with 50 mm supplemental irrigation at sowing, compared to rainfed crop sown in 15 October. Wheat grain yield outputs as yield increases and probability levels have been studied over the rainfed sowing.

3. Results and discussion

3.1. *Weather conditions*

In the 1998/1999 and 1999/2000 seasons, there were adequate rainfall for germination in October, following sowing, with a

Table 4 – Average observed and simulated grain yield, above-ground biomass at harvest and total evapotranspiration for all treatments

Year	Treatments		Wheat Yield (t ha^{-1})		Biomass (t ha^{-1})		ET (mm)	
			Observed	Simulated	Observed	Simulated	Observed	Simulated
1998–1999	A_1							
		B_1	2.81	3.04[a]	7.24	7.42[a]	347	319[a]
		B_2	4.89	5.32	11.63	11.82	492	473
	A_2							
		B_1	3.17	3.06	8.57	7.46	369	325
		B_2	4.66	5.32[a]	13.29	11.82[a]	512	478[a]
1999–2000	A_2							
		B_1	3.56	3.51	6.58	8.18	388	367
		B_2	4.84	4.83	9.23	10.74	497	463
	A_2							
		B_1	3.68	3.54	6.99	8.75	403	373
		B_2	5.08	4.83	9.44	10.74	528	468
2000–2001	A_1							
		B_1	2.85	2.76	5.36	8.95	304	312
		B_2	4.89	5.12	9.87	11.39	491	452
	A_2							
		B_1	4.43	4.00	8.39	10.01	348	346
		B_2	5.46	5.43	10.95	12.06	554	455

[a] Calibrated treatments.

total annual amount of 435 and 416 mm, respectively. In the 2000/2001 season, there were inadequate rainfall for seed germination in October, after sowing. The 2000/2001 season was extremely dry, with a total rainfall of 282 mm. Seed emergence was observed before winter without supplemental irrigation.

Mean maximum and minimum temperatures were slightly higher than the long-term average during the 3 years of study. In Ankara, frost begins in December and ends in March. The coolest months are December, January and February. Snow cover during 2–4 months is common, which protects the emerged crop, as it develops before severe winter. Wheat was in dormancy stage in the field experiment during this period.

3.2. *Observed field results*

The observed biomass at harvest, grain yield, and the ET_{total}, used for model calibration and validation are presented in Table 4. In the first (1998/1999) and second (1999/2000) seasons of the experiment, irrigation at sowing had no effect on grain yields, since wheat received adequate rainfall after sowing in October and, thus, both treatments had similar emergence date. Irrigation in spring and in early summer, however, affected grain yields significantly due to higher evapotranspiration needs. Evapotranspiration in the first season ranged from 347 mm in the rainfed treatment to 512 mm in the irrigated plots and from 388 to 528 mm in the second season.

In the third season (2000/2001), irrigation at sowing had significant effect on grain yield because of insufficient effective rainfall for crop emergence in October. Crop emergence under supplemental irrigation occurred one month earlier than that under rainfed conditions. The early emergence had a significant effect on crop development before the dormant period of frost season. Irrigation during spring and early summer also influenced wheat grain yields and biomass. Evapotranspiration of treatments ranged from 304 mm under rainfed to 554 mm under irrigation.

3.3. *Model simulations*

Seasonal ET values of A_1B_1 and A_2B_2 treatments of the 1998/1999 season were used for model calibration and the results for simulated/observed data pairs are presented in Fig. 1a and b. Statistically, CropSyst model predicted better the seasonal ET under SI (A_2B_2) conditions than under rainfed (A_1B_1) conditions. The RMSE values for A_1B_1 and A_2B_2 were 46 and 25% of the average observed seasonal ET and the Willmott index of agreement were 0.89 and 0.97, respectively. The reason for the lower index of agreement under rainfed conditions, as compared to SI conditions, is probably due the cracking clay soil characteristics where higher evaporation losses occur over the dry period from heading to flowering stages of the crop.

Observed biomass at harvest, grain yield, and ET_{total} values were validated for the three growing seasons (1998/1999–2000/2001) and the results are plotted against observed values for each treatment (Fig. 2).

For grain yield, the simulated/observed data pairs were close to the 1:1 line of perfect agreement (Fig. 2a). The coefficient of determination was high (0.91). The statistical analysis (Table 5) confirmed that CropSyst predicted wheat yield reasonably well. Grain yield results showed a RMSE representing 8% of the average observed grain yield and a high Willmott index of agreement (0.98) (Willmott, 1982).

For biomass, the simulated/observed data pairs were close to the 1:1 line (Fig. 2b), although overestimation occurred in general, with no clear explanation. The regression coefficient was lower (0.64) and the RMSE represented 21% of the

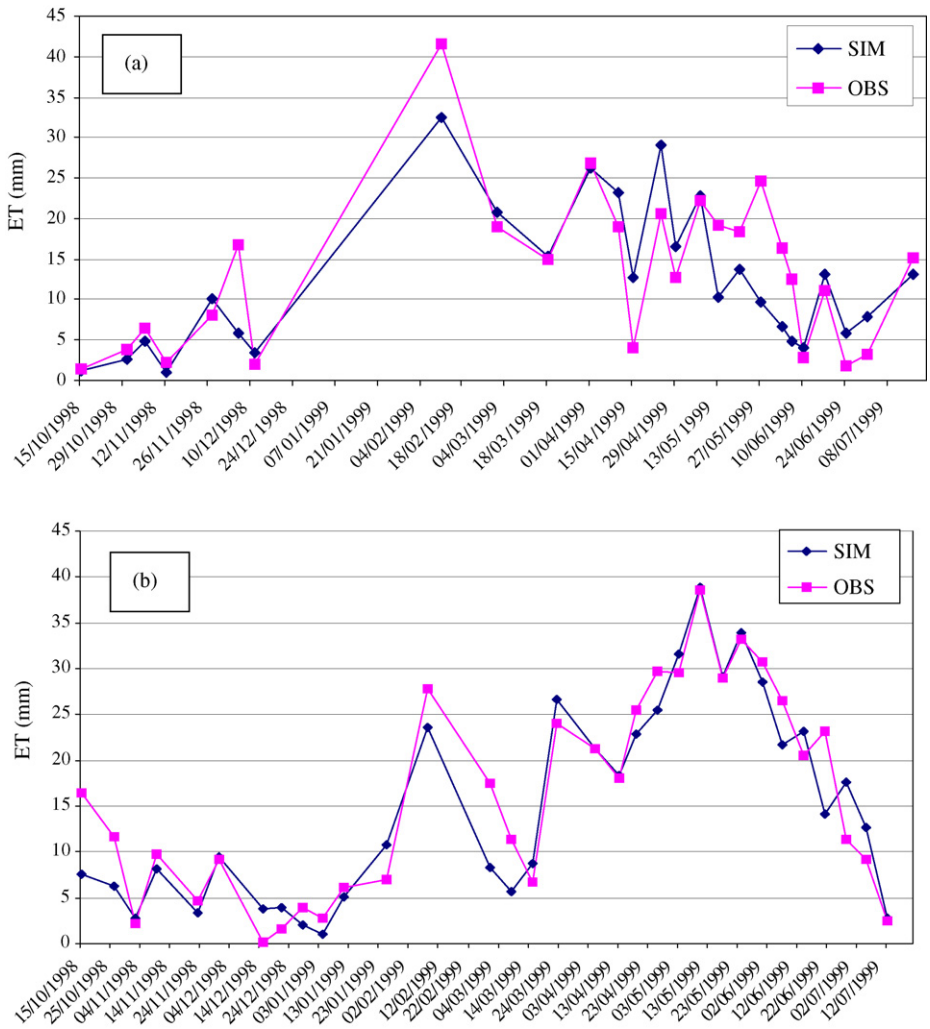

Fig. 1 – Simulated and observed seasonal evapotranspiration for the calibrated year (1998/1999). (a) Rainfed (A1B1), (b) supplemental irrigation (A2B2).

observed average. The Willmott index of agreement between the observed and simulated biomass was 0.76, reasonably close to 1.

The cumulative ET results at harvest also showed most data pairs to be close to the 1:1 line, with a trend to under-prediction (Fig. 2c), and a high coefficient of determination of 0.92, similar to grain yield. The statistical analysis (Table 5) confirmed that CropSyst predicted ET reasonably well. The RMSE values were 11% of the average observed ET and the Willmott index of agreement (0.91) was reasonably close to 1. Similarly, under-prediction of total ET may be attributed to the same reason.

3.4. *Model application scenarios*

The wheat grain yield results for the 20 years (1982–2001) are summarized in Fig. 3. In Ankara, the annual rainfall amount was variable during the last 20 years where wheat yields followed the same trend as that of the annual mean rainfall in the rainfed zone. In the years 1985, 1993 and 2000 inadequate

Table 5 – Summary of statistical results for model validation

Data form	n	Slope	Constant	r^2	Observed mean (t ha^{-1})	Simulated mean (t ha^{-1})	RMSE (t ha^{-1})	RMSE/observed mean	d[a]
Grain yield	10	1.06	−0.32	0.95	4.28	4.24	0.24	0.06	0.98
Biomass	10	0.65	4.38	0.63	8.70	10.01	1.83	0.21	0.76
ET_{total}	10	0.73	85.62	0.91	437[b]	403[b]	47.00[b]	0.11	0.91

[a] d: Willmott index.
[b] ET values are in mm.

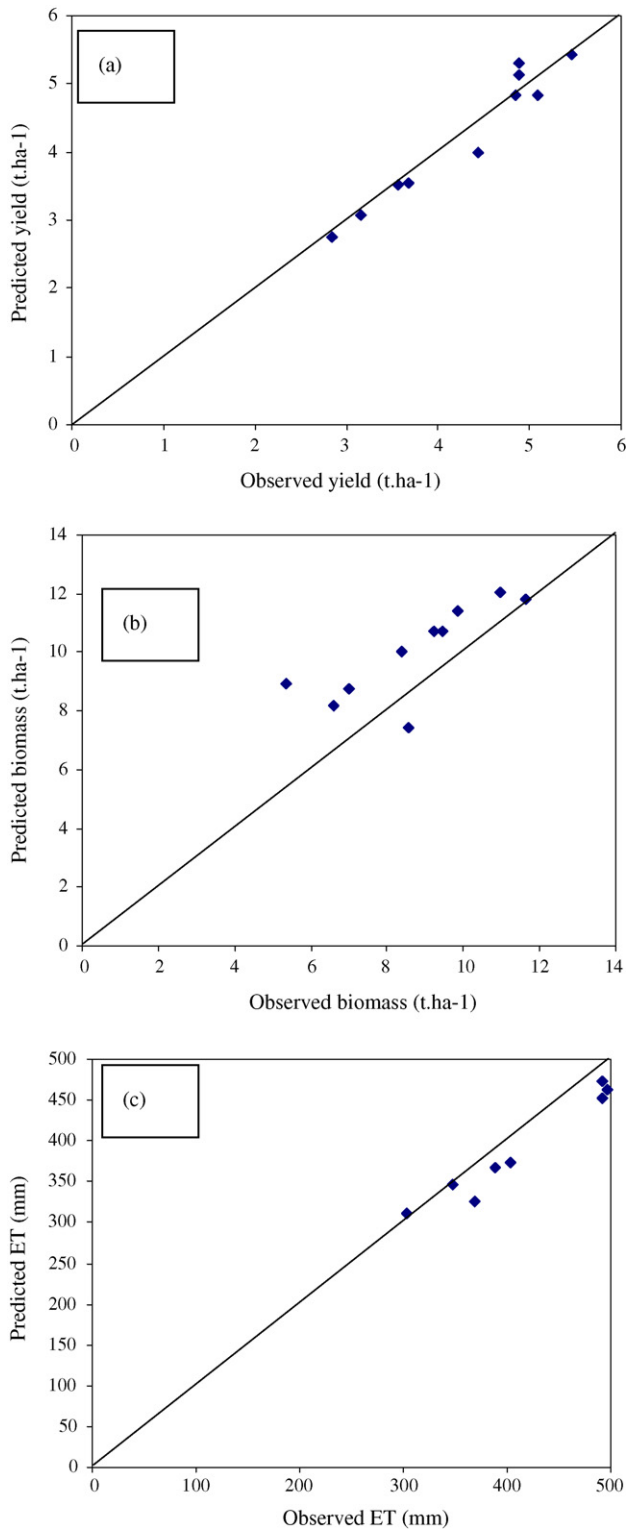

Fig. 2 – Simulated data plotted against observed data for Bezostia wheat cultivar from 1998/1999 to 2000/2001 experimental period. (a) Grain yield (t ha^{-1}), (b) aboveground biomass (t ha^{-1}), and (c) cumulative evapotranspiration (mm).

rainfall occurred, with an amount less then 250 mm, therefore rainfed yields remained very low. However, the rainfall amount was high during the years 1986, 1987, 1994 and 1997 with amount more than 400 mm and resulted in relatively high yields, except 1994 (3.17 t ha^{-1}). The 10 day-dry period (no rainfall) right after the early sowing dates (ESI_1 and ESI_2) for most of the years provided the impact of 50 mm SI over the rainfed wheat yields. Except from 1997 and 1998, where the relatively higher seasonal rainfall there was also adequate rainfall after the sowing (mainly in 1998). Variations in wheat grain yields between sowing dates is most probably coming from other factors like temperatures than rainfall alone.

In dry years, rainfed yields were very low compared to that SI for early sowing. The results, as yield increased over rainfed are summarized in Table 6. The mean increases over rainfed yields were 28, 19 and 13% for the ESI_1, ESI_2 and ESI_3 scenarios, respectively. The highest absolute value was obtained in the year 2001 (111%) by ESI_1 due to the extreme dry period (rainfall of 0.7 mm only) between 15 September and 31 October. Early sowing with SI allowed early crop emergence and development of good stand before being subjected to the winter frost. As discussed before, in years 1997 and 1998 adequate amount of early rain allowed emergence and good crop establishments with optimum growth before the winter cold in December. Therefore, high plant vigour combined with relatively higher rainfall rendered 50 mm SI at sowing not effective. However, overall results show that wheat grain yield could be improved more than 25%, 15% by only 50 mm SI at sowing by ESI_1 in particular, over the rainfed sowing in 15 October. Similar impact of early sowing with SI was also reported in the highland environment of Northwest Iran (Tavakkoli and Oweis, 2004) and in Morocco by Heng et al. (2007).

In central Anatolian plateau, the optimal sowing period extends from the last week of September to mid-October, but probability of sufficient rainfall before mid-October is only

Table 6 – Yield increase (%) by various early sowing supplemental irrigation over rainfed yields for 20 years in Central Anatolia Plateau

	ESI_1	ESI_2	ESI_3
1982	19	18	9
1983	15	10	0
1984	58	45	40
1985	23	11	1
1986	8	4	4
1987	25	17	0
1988	35	33	31
1989	23	10	0
1990	29	19	12
1991	11	6	2
1992	9	7	0
1993	51	39	43
1994	28	11	0
1995	35	13	0
1996	14	7	2
1997	0	0	0
1998	2	1	1
1999	22	14	1
2000	41	24	45
2001	111	89	77

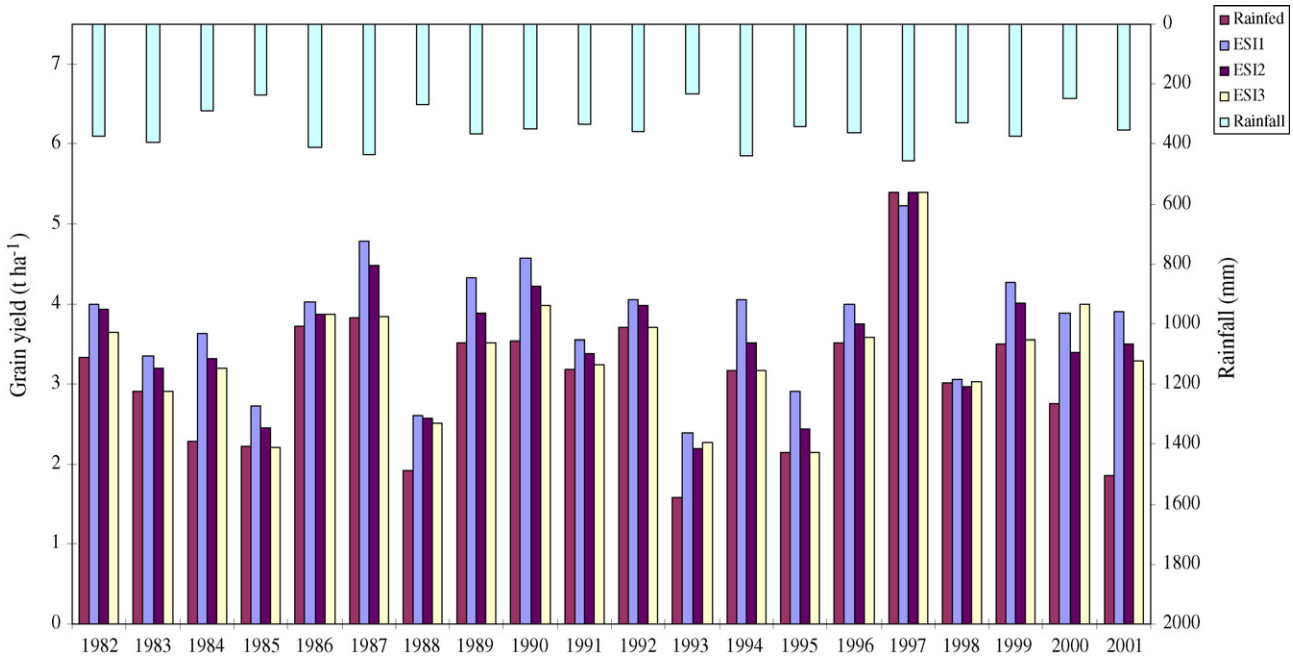

Fig. 3 – Effects of different sowing dates with 50 mm supplemental irrigation on winter wheat grain yield using 20 years climate data.

3.4% increasing to 7, 15 and 20%, by the 20, 25 and 30 of October, respectively. Thus, at least in 3 out of 4 years the expected rainfall will not be early enough or sufficient enough to achieve optimal crop germination and yield. This also means that SI can be effective at sowing in most years. Thus, the important thing here is to apply some water to get early crop emergence to stand cold winter thereafter (Ilbeyi et al., 2006). Modeling applications for several years provided how much yield would be obtained under early sowing with SI to increase the yield stability with reduced risk to help farmers to improve their welfare as well. The amount of rainfall received during the growing season was greater for the early sowing date scenarios; ESI1 and ESI2. Therefore, early planting was more efficient in utilizing water. Keeping this in mind, we have applied the Weibull formula [$P = m/(n + 1)$]) where m is the rank of a value in a list ordered by descending magnitude, and n is the number of years, to determine the probability of the wheat grain yield depending on the different sowing dates with SI (Fig. 4). The results show that in 80% of the years 2.16 t ha^{-1} can be obtained under for rainfed conditions compared with 2.91 t ha^{-1} from ESI1., 2.73 t ha^{-1} from ESI2 and 2.57 t ha^{-1} from ESI3 applications. Moreover, at about 20% of the cases, we obtain 3.70 t ha^{-1} for rainfed compared with 4.33 t ha^{-1} from ESI1, 4.02 t ha^{-1} from ESI2 and 3.98 t ha^{-1} from ESI3

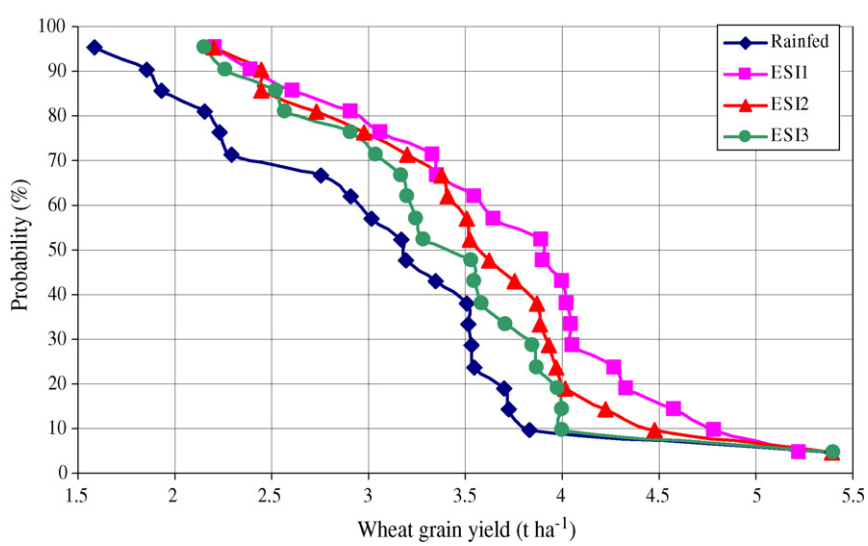

Fig. 4 – Probability of distribution of winter wheat grain yield under different sowing dates with supplemental irrigation (50 mm) compared to rainfed system.

applications. Similar yield reductions were observed in WANA region by Oweis et al. (1998).

4. Conclusion

Calibration of crop input parameters for rainfed and full SI treatment in the season (1998/1999) allowed CropSyst to perform satisfactorily in mimicking grain yield, aboveground biomass, and ET during the three cold winter growing seasons in the highlands of Turkey. In this area, crop emergence, before the severe winter starts, is important to ensure an adequate stand establishment. Thus, early sowing combined with sufficient amount of water as SI, since the probability of sufficient rainfall at this time is very low, would allow crop to get stronger stand before the cold winter starts by about Mid November. Otherwise wheat crop planted late would not emerge because of lower temperatures prevailing at late planting irrespective of higher rainfall obtained. The other years results were then confirmed through modeling as we could not conduct so many years of experiments to see the yield sustainability with reduced risk to recommend farmers a water management strategy to shift the irrigation from later stages of growth in spring-summer months to planting period of Late September to Mid October to utilize the relative optimum air temperatures for optimum crop growth achieving improved water-use efficiency as well as higher biomass and grain yields.

Therefore, these site and year-specific results should be extrapolated in time (long-term responses) and to other regions in order to be more useful. Results from the study suggest that the use of CropSyst after proper calibration and validation can be feasible for such extrapolations.

Given the results reported here, it is expected that the application of the model in similar climates of the region will allow long-term extrapolations with less experimental results, thus reducing time and research cost. Different earlier sowing scenarios in longer period of time (over 20 years) has proved that advancement of sowing date to 15 September compared with rainfed sowing around mid October combined with sufficient water for crop emergence would increase the crop productivity by over 25% and eventually the water productivity as well.

Acknowledgements

The authors wish to thank Drs. Haluk Ustun, Adem Ilbeyi, and their assistants at the Rural Services Research Institute of Ministry of Agriculture, Ankara. Turkey. The authors also wish to thank Drs. Adriana Bruggeman and Ahmed Hachum for their valuable comments and suggestions on the manuscript.

REFERENCES

Cabelguenne, M., Jones, C.A., Marty, J.R., Dyke, P.T., Williams, J.R., 1990. Calibration and validation of EPIC for crop rotations in southern France. Agric. Sys. 33, 153–171.

Heng, L.K., Asseng, S., Mejahed, K., Rusan, M., 2007. Optimizing wheat productivity in two rain-fed environments of the West Asia-North Africa region using a simulation model. Eur. J. Agron. 26, 121–129.

Ilbeyi, A., Ustun, H., Oweis, T., Pala, M., Benli, B., 2006. Wheat water productivity and yield in a cool highland environment: Effect of early sowing with supplemental irrigation. Agric. Water Manage. 82, 399–410.

Kropff, M.J., van Laar, H.H., Matthews, R.B., 1994. ORYZA1, an eco-physiological model for irrigated rice production. In: SARP Research Proceedings, DLO-Research Institute for Agrobiology and Soil Fertility, Wageningen, WAU-Department of Theoretical Production Ecology, Wageningen. International Rice Research Institute, Los Banos, p. 110.

Lengnick, L.L., Fox, R.H., 1994. Simulation by NCSWAP of seasonal nitrogen dynamics in corn: II. Corn growth and yield. Agron. J. 86, 176–182.

Oweis, T., 1997. Supplemental Irrigation: A Highly Water-Efficient Practice. ICARDA, Aleppo, Syria, p. 16.

Oweis, T., Pala, M., Ryan, J., 1998. Stabilizing rainfed wheat yields with supplemental irrigation and nitrogen in a Mediterranean climate. Agron. J. 90, 672–681.

Oweis, T., Salkini, A., Zhang, H., I˙lbeyi, A., Üstün, H., Dernek, Z., Erdem, G., 2001. Supplemental Irrigation Potential for Wheat in The Central Anatolian Plateau of Turkey. ICARDA, Aleppo, Syria, v + 37 pp.

Pala, M., Stockle, C.S., Harris, H.C., 1996. Simulation of durum wheat (*Triticum turgidum* ssp Durum) growth under different water and nitrogen regimes in a Mediterranean environment using CropSyst. Agric. Sys. 51 (2), 147–163.

Perrier, E.R., Salkini, A.B., 1991. Supplementary irrigation in the Near East and North Africa. In: Proceedings of a Regional Consultation on SI. ICARDA and FAO, Rabat Morocco, 7–9 December 1987. Kluwer Academics Publishers.

Raats, P.A.C., Warrick, A.W., 1983. Design and Operation of Farm Irrigation Systems. The American Society of Agricultural Engineers, pp. 125–145.

Stockle, C.O., Martin, S.A., Campbell, G.S., 1994. CropSyst, a cropping systems simulation model: Water and nitrogen budgets and crop yield. Agric. Sys. 46, 335–359.

Stockle, C.O., Nelson, R., 1994. Cropping Systems Simulation Model Users Manual (Version 1. 02. 00) Biological Systems Engineering Department, Washington State University, p. 167.

Tavakkoli, A.R., Oweis, T., 2004. The role of supplemental irrigation and nitrogen in producing bread wheat in the highlands of Iran. Agric. Water Manage. 65, 225–236.

van Keulen, H., Seligman, N.G., 1987. Simulation of Water Use, Nitrogen Nutrition and Growth of a Spring Wheat Crop. Pudoc, Wageningen, p. 310.

Willmott, C.J., 1982. Some comments on the evaluation of model performance. Bull. Am. Meteor. Soc. 63, 1309–1313.